Cunxin Huang

PhD Student | Benchmarking / Evolving Systems / AI for Science

Phone/WeChat: +86 17751373099 Email: huangcunxin@outlook.com GitHub/Web: OptHuang / opthuang.github.io



Profile

- Seeking roles in **AI training and applications**, focusing on AI for Science, algorithm evolution, and benchmarking.
- Knowledge of LLM fundamentals and agent frameworks; practical experience in **tool use, code evolution, and benchmarking**.

Education

2022 – Present • **The Hong Kong Polytechnic University, Applied Mathematics, PhD Student**
Research: derivative-free optimization; supervisors: Prof. Zaikun Zhang, Prof. Xiaojun Chen; **HKPFS** awardee.

2018 – 2022 • **Jilin University, School of Mathematics, BSc**
Information and Computing Science (Computational Mathematics); **rank: 2/109**.
Selected for the "Qiushi Siyuan" Student Development Program (7th cohort).

Internships

Aug 2026 – Present • **Tencent Hunyuan, Post-Training / Eval Team Intern**
- Built a scientific task generation pipeline for SFT/RL post-training data synthesis.
- Plan to lead an operations research benchmark evaluating LLMs' modeling and problem-solving abilities.

May 2026 – Aug 2026 • **Yuanmu Intelligence, Algorithm Intern (Industrial AI / Production Scheduling)**
- Built a heuristic learning framework for industrial algorithms, with benchmarks and evaluators.
- Refactored scheduling operators and evaluation interfaces for agent-driven algorithm adaptation.

Skills and Experience

AI • Agent/RAG, tool use, heuristic learning, code evolution, benchmark/evaluator systems

Engineering • Python, MATLAB, Git/Linux; derivative-free and numerical optimization; Chinese and English

Projects

2026 – Present • **Terminal-Bench Science: Scientific Computing Benchmark for Agents**
[Website / Leaderboard] [GitHub] [v0.1.0]
- **v0.2 Reviewer Group member**; responsible for private-tbs evaluation repository development.
- **Top contributor by task count in v0.1**; **4 co-authored tasks** in optimization, localization, and simulation.
- The benchmark was used in the launch evaluations of **OpenAI GPT-6 Astra** and **Anthropic Claude Fable 5.1**.

• **NextEval: Agent-Based Derivative-Free Optimization and Benchmarking** [Website]
- Combined agents and numerical methods for tool selection, strategy adaptation, and budget allocation.
- Designed NextEval Bench with common budget and information constraints for models, frameworks, and solvers.

• **OptiProfiler Evolve: Automated DFO Solver Evolution**
- Built a solver evolution system using OptiProfiler, with staged benchmarking and held-out validation.

2024 – Present • **OptiProfiler: DFO Benchmarking and AI Agents** [Platform] [GitHub] [Agent]
- Primary developer; built DFO benchmarking and agent tools for cloud execution, analysis, and diagnostics.

Selected Research

2026 • **NextEval: An Agent-Based Solver and Benchmark Protocol for Derivative-Free Optimization** (submitted to ICLR)
• **OptiProfiler: a platform for benchmarking optimization solvers** (with T. M. Ragonneau and Z. Zhang)
• **Non-convergence Analysis of Probabilistic Direct Search** [arXiv:2606.01320] (with Z. Zhang)
• **A Unified Series Condition for the Convergence of Derivative-Free Trust Region and Direct Search** (with Z. Zhang)

2023 • **Error Analysis of Finite Difference Scheme for American Option Pricing** (*J. Comput. Appl. Math.*)

Honors and Recognition

2026 • **SIAM OP26 Student Travel Award**; invited talk at SIAM OP26

2024–2025 • ICCOPT / ISMP invited talks; ORSC Application Award (team member); PolyU Best Presentation Award

2021–2023 • ORSC Math Programming Forum: Outstanding Presentation; MCM/ICM Finalist (top 2%); Qi An-Xin Scholarship

2019–2022 • **Hong Kong PhD Fellowship Scheme (HKPFS)**; National Scholarship; Zhao-Yue Scholarship

References

- **Michael A. Saunders**, Professor (Research) Emeritus, Stanford; SIAM Fellow [Website]
- **Philippe L. Toint**, Emeritus Professor, University of Namur; SIAM Fellow; former MOS Chair [Website]